A bounded rational solution of the homogeneous edge system yields a linearly ordered colouring with two plus log-two of the dynamic range many colours

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Allowed standard prerequisites (editorial).

- The linearly ordered (LO) colouring of a hypergraph: a colouring in which every edge has a unique maximum colour. Defined in the introduction of the paper, before source line 100; the counted proof recalls the equivalent

form in which the minimum colour of each edge is unique. Fixes the property that the dyadic colour assignment in the proof has to satisfy.

- The homogeneous system Av = 0^m attached to the edge set, and its solvability from LO 2-colourability. Set up in the section of which the counted lemma is the first statement (source lines 240-265), where the author also notes that every coordinate of the reference solution w equals -1/3 or 2/3. Supplies the solution u whose minimal and maximal absolute coordinate values M and m the colour bound 2 + log_2(M/m) is stated in terms of.

Lemma 1. *Suppose we have a solution, u, to the homogeneous system where M is the maximal value of the absolute value of a coordinate and $m > 0$ is the minimal absolute value. Then, we can LO colour with $2 + \log_2(M/m)$ colours.*

Proof. For notational convenience let us instead require that the minimal colour in each edge should be unique. We can simply reverse the order of the colours at the end. By scaling we can assume $M = 1$. We use even colours for positive coordinates and we give x the colour 2ℓ if v_x is at most $2^{-(2\ell-1)}$ and strictly larger than $2^{-(2\ell+1)}$. For negative coordinates we use $2\ell + 1$ as the colour if v_x is between $-2^{-2\ell}$ (inclusive) and $-2^{-(2\ell+2)}$ (non-inclusive). Let us verify that this gives a correct colouring.

Take an edge (x, y, z) and suppose both x and y get the same colour 2ℓ. Then by the linear equation of the edge $v_z < -2^{-2\ell}$ and thus z has a colour below 2ℓ. The case of two vertices of odd colour is similar and as the bound on the number of colours is immediate, the lemma follows. □